Interview RAG Assistant – Problem Statement

Problem Statement

Technical interview preparation has become increasingly challenging due to the vast amount of learning material available across different platforms such as PDFs, notes, blogs, online courses, documentation, and interview experiences. Candidates often spend significant time searching for relevant information instead of focusing on learning and practice. Traditional search methods require users to manually browse through multiple documents, which can be time-consuming and inefficient. Furthermore, general-purpose AI chatbots may generate answers that are not based on the user's study materials, leading to inaccurate responses, missing context, or hallucinations. There is a growing need for an intelligent interview preparation system that can understand user queries, retrieve the most relevant information from trusted resources, and provide accurate, context-aware answers. Such a system should reduce the effort required to find information, improve learning efficiency, and help candidates prepare more effectively for technical interviews.

Proposed Solution

The proposed solution is a Retrieval-Augmented Generation (RAG) based Interview Assistant designed to provide reliable and personalized interview support. The system allows users to upload interview preparation resources such as PDFs, text documents, notes, and study guides. The uploaded content is processed through text extraction, chunking, and embedding generation, and then stored in a vector database for efficient semantic retrieval. When a user asks a question, the system first retrieves the most relevant document chunks based on meaning rather than simple keyword matching. These retrieved contexts are then supplied to a Large Language Model (LLM), enabling it to generate accurate, context-rich responses grounded in the user's own knowledge base. By combining information retrieval with generative AI, the system minimizes hallucinations, improves answer quality, and provides a more trustworthy interview preparation experience. The assistant can be used for concept clarification, interview question practice, revision, and quick knowledge retrieval across large collections of learning materials.

Objectives

- Improve interview preparation efficiency through intelligent document retrieval.
- Provide accurate and context-aware answers based on uploaded resources.
- Reduce hallucinations commonly found in generic AI chatbot responses.
- Enable semantic search across large interview knowledge repositories.
- Support personalized learning and interview revision workflows.
- Enhance candidate confidence and readiness for technical interviews.

Expected Outcome

The final system will function as a web-based AI Interview Assistant capable of understanding natural language questions, retrieving relevant information from a custom knowledge base, and generating accurate answers in real time. The solution will help users quickly access important concepts, strengthen interview preparation, and improve overall learning productivity.